

Smart Traffic Lights Using Local AI for Adaptive Traffic Management and Emergency Vehicle Priority

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Abstract—Traditional traffic lights commonly operate using fixed timing plans and therefore have limited ability to respond to rapidly changing traffic conditions. Recent research has investigated adaptive traffic signal control using artificial intelligence, machine vision, and reinforcement learning. This paper presents a conceptual smart traffic-light architecture in which roadside cameras provide traffic information to local computing units, allowing traffic conditions to be processed near the intersection. The architecture also considers an emergency-vehicle priority mechanism so that nearby traffic signals can coordinate a route for an ambulance or other emergency vehicle. The proposed approach is intended to reduce dependence on continuous cloud processing and provide a framework for responsive traffic management. The paper summarizes relevant literature, presents the proposed architecture, identifies the research gap, and discusses directions for future implementation and evaluation.

Index Terms—smart traffic lights, local AI, edge computing, machine vision, traffic signal control, emergency vehicle priority

I. INTRODUCTION

Urban traffic conditions can change significantly over short periods. A fixed-time traffic signal uses predetermined signal durations and therefore cannot directly react to sudden changes in traffic demand. Artificial intelligence (AI) and adaptive traffic signal control have been investigated as approaches for making signal decisions from traffic information. A review by Rasheed et al. describes the application of deep reinforcement learning to traffic signal control and discusses its potential for addressing complex and changing traffic conditions [1].

Machine-vision approaches provide another way to obtain traffic information. Gorodokin et al. investigated adaptive traffic-light control using machine vision and vehicle-flow parameters such as vehicle counts, queue information, and vehicle dynamics [2]. These studies motivate a system in which traffic conditions are measured continuously and signal timing is adjusted according to the observed road state.

This paper proposes a conceptual architecture called “Smart Traffic Lights Using Local AI.” The main idea is to process traffic-camera information close to the intersection instead of sending all raw video to a remote cloud server. The architecture also includes an emergency-vehicle priority function, because traffic-signal preemption has been studied as a method for helping emergency vehicles pass congested intersections [3].

II. RELATED WORK

A. AI-Based Traffic Signal Control

Rasheed et al. reviewed deep reinforcement learning (DRL) methods for traffic signal control. Their review covers traffic signal control characteristics, DRL architectures, simulation platforms, complexity, design considerations, and open research issues [1]. The work shows that learning-based traffic control is an active research area.

B. Machine Vision for Adaptive Signals

Gorodokin et al. proposed an approach based on machine vision for adaptive traffic-light control. Their study used camera-based traffic information and considered vehicle-flow parameters, queue conditions, and vehicle dynamics when determining traffic-light phases and cycle durations [2]. This supports the use of cameras as an information source for adaptive signal control.

C. Emergency Vehicle Signal Control

Emergency vehicle priority is an important extension of adaptive traffic management. Cao, Shuai, and Li studied emergency-vehicle-centered traffic signal control and discussed the need for appropriate signal preemption when emergency vehicles encounter congested intersections [3]. Their work provides a research basis for incorporating emergency-vehicle priority into a smart signal architecture.

III. PROPOSED SYSTEM

The proposed system consists of four main stages:

- 1) **Traffic sensing:** Roadside cameras capture information about vehicles and traffic conditions.
- 2) **Local processing:** A smart roadside computer processes the camera information locally and estimates the traffic state.
- 3) **Decision and coordination:** The local controller determines an appropriate signal action. Selected information may also be forwarded to a city server for storage and wider analysis.
- 4) **Traffic-light control:** The controller sends the required command to the traffic-signal controller.

The conceptual information flow is:

*Street Cameras → Local Smart Computer → Traffic
Decision → Traffic Lights*

For city-level monitoring, an optional connection can be added:

Local Smart Computer → City Cloud Server

The local-processing concept is intended to reduce the need to transmit continuous raw camera information to a remote server. The exact latency, cost, and network savings must be measured experimentally rather than assumed.

IV. EMERGENCY VEHICLE PRIORITY

The proposed architecture can be extended with an emergency-vehicle detection module. When an authorized ambulance or other emergency vehicle is detected, the system can coordinate nearby traffic signals to provide an appropriate priority phase. This concept is related to emergency vehicle preemption research reported by Cao et al. [3].

A possible control sequence is:

- 1) Detect and verify an emergency vehicle.
- 2) Determine the vehicle's approach direction and intended intersection.
- 3) Check the current signal state and traffic conditions.
- 4) Coordinate a safe priority phase.
- 5) Return the intersection to normal adaptive operation after the emergency vehicle passes.

The exact detection method, communication mechanism, safety constraints, and signal-transition rules would need to be defined and validated before real-world use.

V. RESEARCH GAP

Existing research demonstrates the use of machine vision and AI for adaptive traffic signal control, while separate work has investigated emergency-vehicle signal preemption [1]–[3]. A practical student-level system can therefore focus on integrating these ideas into a single local-processing architecture.

The proposed research gap is the development and evaluation of a low-cost prototype that combines local traffic-state estimation with emergency-vehicle priority while keeping the design simple enough for deployment at an individual intersection. Important evaluation questions include detection accuracy, decision latency, signal-transition safety, communication requirements, and the effect on both emergency-vehicle delay and ordinary traffic.

VI. SYSTEM EVALUATION PLAN

The prototype can be evaluated using the following measurable parameters:

- vehicle-detection accuracy;
- traffic-density estimation accuracy;
- emergency-vehicle detection accuracy;
- decision latency at the local controller;
- emergency-vehicle waiting time;
- average waiting time for ordinary vehicles; and
- network data transmitted by the local-processing system.

These measurements should be obtained experimentally using a controlled simulation or test setup. Numerical performance values should not be claimed until they have been measured.

VII. DISCUSSION

Local processing can be useful when a traffic application requires rapid decisions from camera data. Machine-vision research demonstrates that traffic information can be extracted from cameras and used to adapt signal operation [2]. AI-based traffic-signal research also indicates that learning-based methods can be applied to complex traffic-control problems [1].

However, a practical deployment must address several limitations. Camera performance can be affected by weather, lighting, occlusion, and camera placement. Emergency-vehicle detection also requires reliable identification and safeguards against false priority requests. In addition, any change to traffic-signal operation must follow applicable traffic-safety requirements and should be tested before real-world use.

VIII. CONCLUSION

This paper presented a conceptual smart traffic-light architecture based on local AI processing and adaptive traffic information. The architecture uses roadside cameras, local processing, and traffic-signal control, with an optional cloud connection for storage and wider analysis. The design also incorporates emergency-vehicle priority as an extension of adaptive traffic management. Existing literature supports the use of AI, machine vision, and emergency-vehicle signal preemption in traffic-control research [1]–[3]. Future work should implement the prototype and experimentally measure detection accuracy, latency, traffic delay, and emergency-vehicle response.

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